



Revealing that the Gas-phase Reaction of CN with H₂S Can Be a Source of HSCN in Interstellar Molecular Clouds

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Abstract

Sulfur-bearing molecules are key constituents of the interstellar medium (ISM). Particularly, hydrogen sulfide (H₂S) and cyano (CN) radicals are key precursors of prebiotic molecules in the ISM. However, the ultralow-temperature gas-phase reactivity remains poorly characterized yet. We report the first experimental and theoretical investigation of the CN + H₂S reaction under conditions relevant to cold molecular clouds. Rate coefficients were determined between 11.7 and 45.5 K using the Cinétique de Réaction en Ecoulement Supersonique Uniforme technique coupled with pulsed laser photolysis–laser-induced fluorescence, yielding negligible temperature dependence values around $4.0 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$ in excellent agreement with complementary rate coefficients calculations. AutoMeKin and coupled-cluster theory reveal that the dominant channel involves CN addition to H₂S, followed by H elimination, forming HSCN. This pathway is energetically more favorable than the previously assumed HCN + SH channel and exhibits submerged transition states, suggesting efficient reactivity at ultracold temperatures. Astrochemical modeling indicates that inclusion of this reaction in chemical networks enhances HSCN abundances in dark clouds, with contributions comparable to those from dissociative recombination routes. Although the CN + H₂S reaction is absent from current astrochemical databases, our results demonstrate its potential role in sulfur–nitrogen coupling and the formation of prebiotic molecules in the ISM. These findings underscore the need to update chemical models to account for this process and improve predictions of sulfur chemistry in star-forming regions.

Unified Astronomy Thesaurus concepts: Dense interstellar clouds (371); Pre-biotic astrochemistry (2079); Laboratory astrophysics (2004); Chemical kinetics (2233)

1. Introduction

Hydrogen sulfide (H₂S) is the simplest sulfur-containing molecule and plays a fundamental role in the global sulfur biogeochemical cycle on Earth. It was particularly crucial during the origin of life on our planet, as volcanic emissions of H₂S contributed to the prebiotic synthesis of sulfur-containing amino acids such as cysteine and methionine (N. M. Kredich 2008; E. T. Parker et al. 2011). Nowadays, H₂S is commonly found in soils and is produced by anaerobic bacteria in wetlands, salt marshes, and estuarine ecosystems, where it is responsible for the characteristic odor of rotten eggs. In the Earth's atmosphere, H₂S acts as a pollutant with a relatively short lifetime, primarily due to its rapid oxidation by hydroxyl (OH) radicals, forming sulfur dioxide (SO₂), with an effective lifetime of approximately 2 days (R. Atkinson et al. 2004).

Beyond Earth, H₂S has also been detected in the interstellar medium (ISM) both in ice and gas phases. In 1972, the first observations of H₂S were reported toward several warm interstellar clouds such as Orion A, W51, and DR21(OH) (P. Thaddeus et al. 1972) but it has also been identified in other astrophysical environments, including cold dark clouds (Y. C. Minh et al. 1989), circumstellar envelopes around evolved stars (N. Ukita & M. Morris 1983), diffuse clouds (R. Lucas & H. S. Liszt 2002), and protoplanetary disks (N. T. Phuong et al. 2018). Another key precursor of prebiotic molecules is the cyano (CN) radical. CN radicals are widespread in the ISM, having been detected in diffuse clouds such as ζ Oph (A. McKellar 1940; W. S. Adams 1941), giant molecular clouds such as the Orion Nebula and W51 (K. B. Jefferts et al. 1970), and cold clouds (E. Churchwell 1980). Thus, CN and H₂S coexist in many environments, including cold interstellar clouds.

To our knowledge, the gas-phase reaction between CN and H₂S has not been investigated experimentally, and no rate coefficients for Reaction (R1), $k(T)$, have been reported to date.

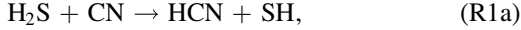


In contrast, theoretical works report that Reaction (R1) may play a pivotal role in the formation of hydrogen cyanide

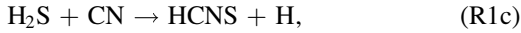
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(HCN), the simplest cyanide compound, and the mercapto (SH) radical via a H-atom abstraction channel (Reaction (R1a)). Although the formation of HNC (Reaction (R1b)) was also postulated by A. D. Burke et al. (2021), the production of HCN is thermodynamically more favorable.



Other potential reaction channels in the $\text{H}_2\text{S} + \text{CN}$ reaction are those involving the addition of CN radical and the subsequent H-elimination (Reactions (R1c)–(R1e)). The relative importance of these channels forming HCNS (thiofulminic acid), HNCS (hydrogen isothiocyanate), and HSCN (thiocyanic acid) has not been investigated yet.



Even though HCNS, HNCS, and HSCN are well-known molecular components of interstellar clouds since 1979, recent detections of HCNS, HNCS, and HSCN in the star-forming nebula known as Taurus Molecular Cloud 1 (TCM-1) through the QUIJOTE spectral line survey suggest that the $\text{H}_2\text{S} + \text{CN}$ reaction may play a key role in their formation via CN-addition plus H-elimination channels (Reactions (R1c)–(R1e); J. Cernicharo et al. 2024). The gas-phase formation mechanism of HNCS and HSCN has been recently revisited by J. Cernicharo et al. (2024), starting with the protonation of NCS by abundant proton donors XH^+ , such as HCO^+ , H_3O^+ , and H_3^+ , and followed by electron transfer reactions. As for HCNS, its formation route is still unclear. According to J. Cernicharo et al. (2024), HSCN is 2.6 times more abundant than HNCS, implying that the formation channel (Reaction (R1e)) is more efficient.

Therefore, we present in this work the first comprehensive study of the $\text{H}_2\text{S} + \text{CN}$ reaction from the kinetic, mechanistic, and astrophysical points of view.

2. Experimental Methodology

2.1. Pulsed CRESU Setup

From the kinetic point of view, $k(T)$ has been experimentally determined under low-temperature conditions (11.7–45.5 K) and total pressures between 0.05 and 0.117 mbar using the Cinétique de Réaction en Ecoulement Supersonique Uniforme (CRESU) technique coupled to the Pulsed Laser Photolysis—Laser Induced Fluorescence (PLP-LIF) kinetic technique. (A detailed description of the technique and the setup can be found in A. Canosa et al. (2016), D. González et al. (2025), E. Jiménez et al. (2015), and A. J. Ocaña et al. (2019)). A short description is given below.

Two different Laval nozzles and a total of four operational conditions were used in this work to fill up the temperature range (see Table 1). The isentropic expansion from a relatively high-pressure small chamber known as *reservoir* to a low-pressure bigger chamber through the Laval nozzle results in a uniform supersonic cold jet. A rotary disk with one or two symmetrical apertures was employed to pulse the gas at 5 Hz for $T = 11.7$ K or 10 Hz for $T > 11.7$ K.

The PLP-LIF kinetic technique to follow the reaction progress was described in A. J. Ocaña et al. (2019). CN radicals were generated by UV photolysis of gaseous ICN

Table 1

Summary of the Experimental Conditions Employed in This Work (Temperature, Gas Density (n), and $[\text{H}_2\text{S}]$ in the Supersonic Jet) and the Obtained $k(T)$, as a Function of Temperature^a

T/K	$n/10^{16} \text{ cm}^{-3}$	$[\text{H}_2\text{S}]/10^{13} \text{ cm}^{-3}$	$k(T)/10^{-10} \text{ cm}^3 \text{ s}^{-1}$
11.7 ± 0.7	6.88 ± 0.62	0.14–4.18	3.59 ± 0.42
18.6 ± 0.6	2.48 ± 0.32	0.20–2.90	3.68 ± 0.39
21.1 ± 0.5	3.37 ± 0.15	0.11–1.19	4.70 ± 0.76
45.5 ± 2.0	0.69 ± 0.08	0.12–1.31	3.96 ± 0.51

Note.

^a Uncertainties in $k(T)$ account for statistical ($\pm 2\sigma$) and 10% systematic errors. In n , the indicated errors result from aerodynamic fluctuations in the flow ($\pm 1\sigma$).

introduced into the *reservoir* by flowing the buffer gas through powder ICN. The photolysis radiation at 248 nm was generated by a KrF excimer laser (Coherent, model ExciStar XS 200). CN radicals were then excited at 388 nm (0,0 band), radiation achieved with a frequency-doubled dye laser (Lambda Physik, model Scanmate) pumped by the second harmonic of a Nd:YAG laser (Continuum, model Surelite III). The temporal evolution of CN radicals was monitored by LIF (I_{LIF}) at ~ 420 nm, using a photomultiplier tube (Electron Tube, model 9813B) with a bandpass filter centered at 420 nm with a full width at half-maximum of 10 nm (Andover Co., model 420FS10-25).

2.2. Kinetic Method

The kinetic experiments were carried out under *pseudo*-first-order conditions, with $[\text{H}_2\text{S}]$ in large excess with respect to the initial concentration of CN radicals ($[\text{H}_2\text{S}] \gg [\text{CN}]_0$). The temporal profiles of I_{LIF} after rotational relaxation ($t > 20 \mu\text{s}$) are exponential decays mainly due to Reaction (R1). The analysis of I_{LIF} yields the *pseudo*-first-order rate coefficient, k' (or k'_0 in the absence of H_2S), which is proportional to $[\text{H}_2\text{S}]$ as shown by Equation (E1):

$$k' = k(T)[\text{H}_2\text{S}] + k'_0. \quad (\text{E1})$$

At a single temperature, the kinetic measurement was repeated several times, and $k(T)$ is obtained from the slope of the $k' - k'_0$ vs. $[\text{H}_2\text{S}]$ plots (see Figure 1). The linear least-squares fitting was performed considering the standard deviation ($\pm \sigma$) in the $k' - k'_0$ values with a weighting factor of $1/\sigma^2$. It is very important to perfectly know $[\text{H}_2\text{S}]$ in the jet to accurately obtain $k(T)$ at each temperature. Gaseous H_2S was introduced into the *reservoir* through a calibrated mass flow controller from a diluted mixture stored in a Pyrex bulb (mixing ratios, f , between 1.01% and 3.00%). $[\text{H}_2\text{S}]$ were determined by the mass flow rate introduced into the reaction chamber relative to the total mass flow, as explained in M. Antiñolo et al. (2016), A. J. Ocaña et al. (2019), and S. Blázquez et al. (2020).

2.3. Chemicals

Buffer gases (He and N_2) with 99.999% purity (Nippon Gases Europe) were used as supplied. Gaseous H_2S was purchased from Sigma-Aldrich ($>99\%$). Solid ICN was acquired from Acros Organics (98%).

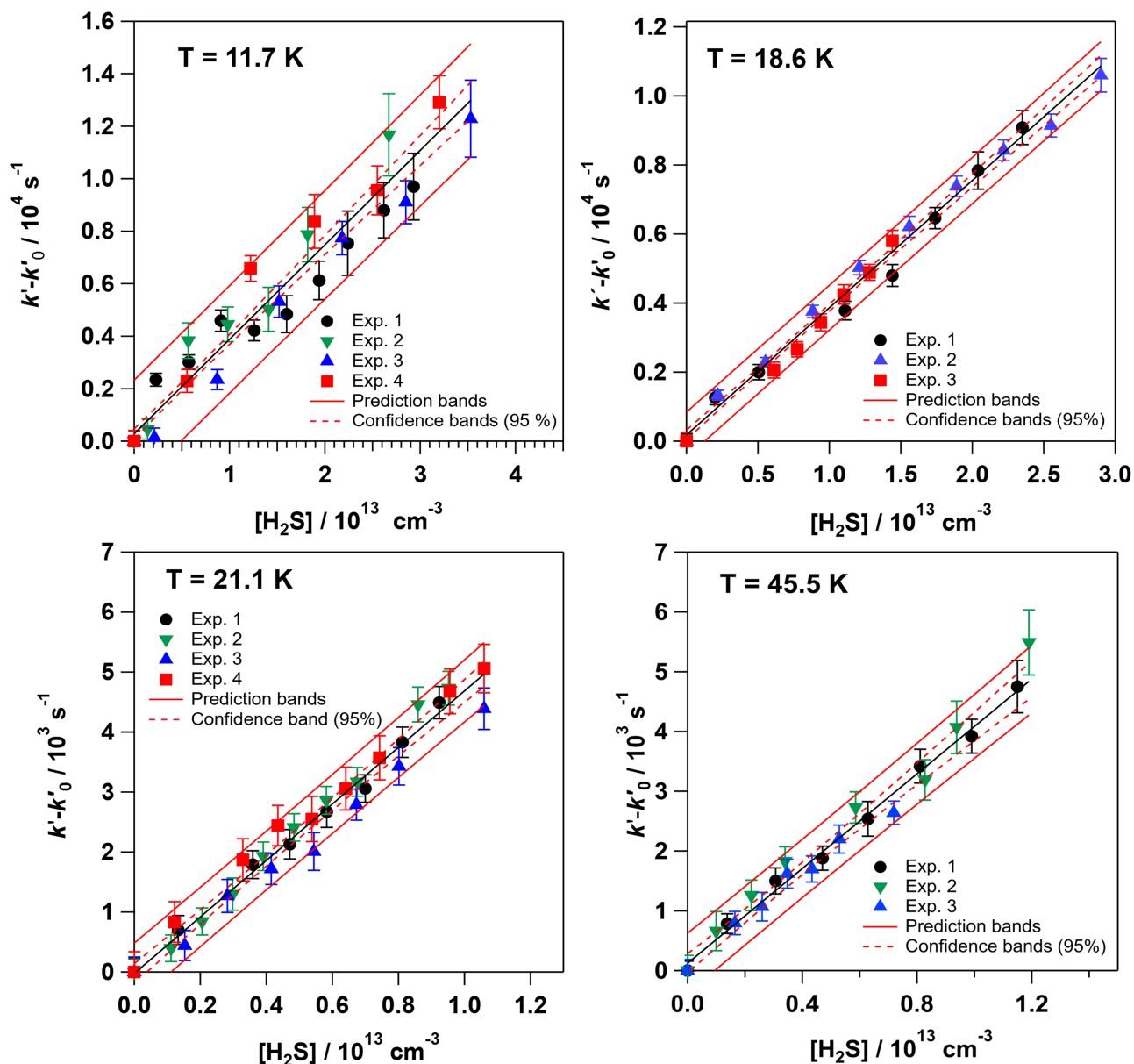


Figure 1. Examples of $k' - k'_0$ versus $[H_2S]$ plots at different temperatures.

3. Computational Methodology

The kinetics of Reaction (R1) was investigated theoretically over the 10–100 K temperature range using coupled-cluster theory at the CCSD(T)-F12-RIJ/aug-cc-pVTZ level. In parallel, the formation mechanisms leading to HCN and HSCN were explored within the CH_2NS chemical space. The associated chemical reaction network (CRN) comprising all [C, H₂, N, S] isomers and their interconversion and reaction pathways was constructed using the AutoMeKin program (E. Martínez-Núñez 2015a, 2015b; E. Martínez-Núñez et al. 2021). AutoMeKin is an open-source platform for the automated exploration of reactive pathways that combines reactive molecular dynamics (MD) simulations with graph-theoretical algorithms and interactive visualization tools, enabling the efficient generation and analysis of complex reaction mechanisms.

The computational protocol initiates with a preliminary exploration of the potential energy surface (PES) employing

semiempirical methods (denoted as Level1), which is subsequently refined at a higher level of theory (Level2) for the accurate characterization of stationary points. In accordance with previous studies, Level1 calculations were carried out using the PM7 method (J. J. P. Stewart 2013), while Level2 geometry optimizations and frequency calculations were performed using density functional theory (DFT) with the B3LYP functional and the 6-31G* basis set. All Level2 computations were executed with the GAUSSIAN09 software suite (M. J. Frisch et al. 2010).

For the two most favorable pathways involving Reactions (R1a) and (R1e), the key stationary points were further optimized using coupled-cluster theory at the CCSD(T)-F12-RIJ/aug-cc-pVTZ level (hereafter referred to as Level3), as implemented in ORCA5.0 (F. Neese 2003, 2023; M. Garcia-Ratés & F. Neese 2020). The AutoMeKin workflow was iterated 40 times, each cycle comprising new sets of MD trajectories, optimization of newly discovered minima and transition states (TSs), and subsequent CRN updates. Each MD batch included 500

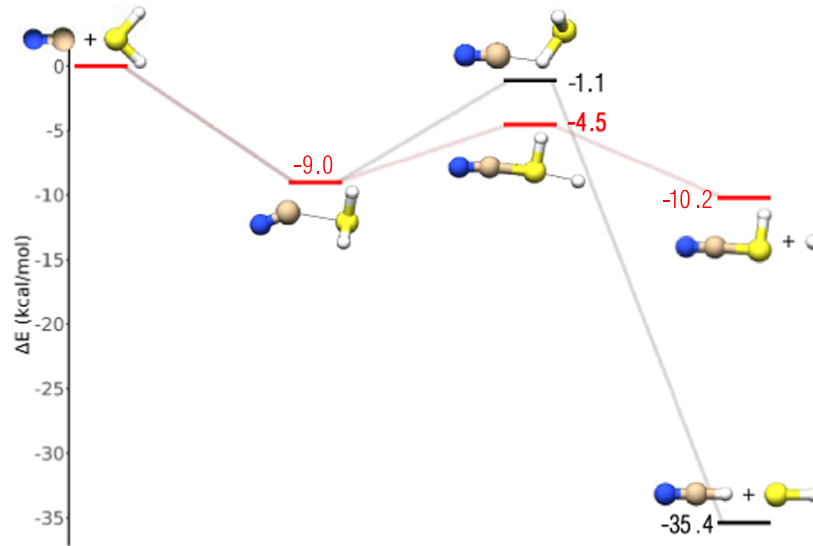


Figure 2. Level3 calculated energy profile for reactions (R1a, black) and (R1e, red) including zero-point vibrational energy (ZPVE) corrections. The reactants (CN + H₂S), the van der Waals intermediate, transition states, and products are shown along each reaction pathway. Atomic color scheme: N = blue, C = beige, S = yellow, and H = white.

trajectories, with a maximum integration time of 0.5 picoseconds per trajectory.

Although this study focuses specifically on the above CN + H₂S reactions, the full reaction network is publicly available via Zenodo (doi:10.5281/zenodo.17749441; E. Martinez-Nunez 2025). Within the CRN files, species are systematically labeled as MIN (minima), TS, or PROD (products), with numerical identifiers used to distinguish among distinct molecular structures. Global rate coefficients were determined using the competitive canonical unified statistical (CCUS; L. G. Gao et al. 2018; D. González et al. 2022) model. According to this model, the rate coefficient for the overall process k^{CCUS} is given by

$$k^{\text{CCUS}} = \left(\frac{1}{k^{\text{assoc}}} + \frac{1}{\sum_{i=1}^2 k_i^{\text{CVT}}} \right), \quad (\text{E2})$$

where $i = \text{R1a}, \text{R1e}$. The association rate coefficient between CN and H₂S, denoted as k^{assoc} , was approximated using the following expression (Y. Georgievskii & S. J. Klippenstein 2005):

$$k^{\text{assoc}} = C\mu^{-1/2}(d_{\text{CN}}d_{\text{H}_2\text{S}})^{2/3}T^{-1/6}. \quad (\text{E3})$$

In this equation, μ denotes the reduced mass of the two reactants (in amu), T is the temperature (in K), and the dipole moments $d_{\text{CN}} = 1.37439$ D and $d_{\text{H}_2\text{S}} = 0.99425$ D were computed at Level3. The constant C has a value of 1.83×10^{-9} , with units such that k^{assoc} is expressed in $\text{cm}^3 \text{s}^{-1}$. In Equation (E2), k_i^{CVT} represents the canonical variational transition state theory (CVT) rate coefficient for reaction i and is given by (B. C. Garrett & D. G. Truhlar 1979a, 1979b)

$$k_i^{\text{CVT}} = \Gamma_i^{\text{CVT}} k_i^{\text{TST}}. \quad (\text{E4})$$

The factor Γ_i^{CVT} accounts for dynamical recrossing effects in reaction i , and was computed at the M06-2X/MG3S level of theory, which provided the best agreement with Level3 among the seven DFT methods tested (see the Appendix for details). The factor k_i^{TST} is the conventional transition state theory rate coefficient, calculated from Level3-calculated structures,

Table 2
CCUS Rate Coefficients for the CN + H₂S → H + HSCN Reaction

T (K)	k^{CCUS} ($10^{-10} \text{ cm}^3 \text{ s}^{-1}$)
10	4.00
20	3.56
30	3.33
40	3.17
50	3.06
60	2.96
70	2.89
80	2.83
90	2.77
100	2.72

energies, and vibrational frequencies. Finally, the individual CCUS rate coefficients for each pathway were obtained from

$$k_i^{\text{CCUS}} = k^{\text{CCUS}} \frac{k_i^{\text{CVT}}}{\sum_{i=1}^2 k_i^{\text{CVT}}}. \quad (\text{E5})$$

4. Results and Discussion

4.1. Potential Energy Surface and Rate Coefficients

Our autonomous exploration of PES revealed a reaction pathway (Reaction (R1e)) leading to the formation of atomic hydrogen and HSCN. Notably, this newly identified pathway is energetically more favorable than the previously reported mechanism (Reaction (R1a)), yielding HCN and SH. Both pathways involve submerged transition states; however, the transition state associated with the formation of HSCN lies 3.4 kcal/mol lower in energy (Figure 2). Table 2 presents the computed global rate coefficients in the 10–100 K temperature range for Reaction (R1). The H₂S + CN → H + HSCN pathway is the predominant channel, being channel (Reaction (R1a)) 6 orders of magnitude slower. Additional details on the calculated rate coefficients are provided in the Appendix.

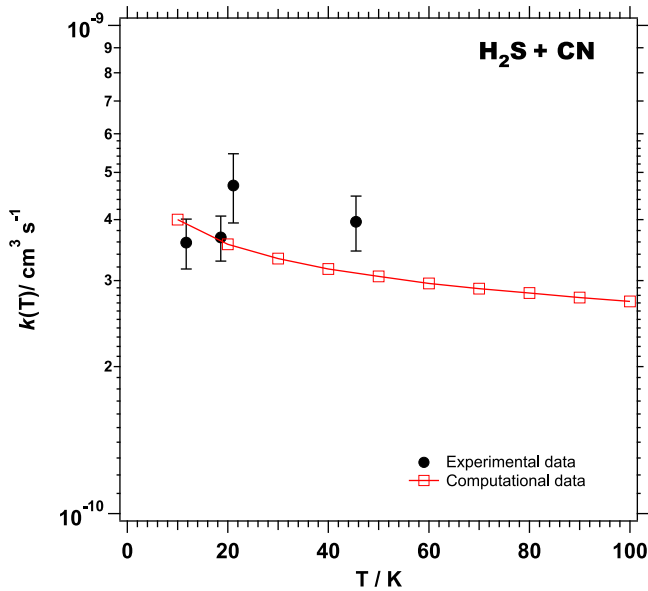


Figure 3. Temperature dependence of the global rate coefficient for the $\text{H}_2\text{S} + \text{CN}$ reaction.

As shown in Table 2, a slight negative temperature dependence is predicted for $k(T)$ between 10 and 100 K, i.e., $k(T)$ increases when T decreases. For this CN reaction, the rate coefficient varies around 30% in that temperature range. Experimental $k(T)$ obtained between 11.7 and 45.5 K are listed in Table 1 together with some experimental conditions during the performance of the kinetic study (gas density (n), $[\text{H}_2\text{S}]$, and temperature). As shown, $k(T)$ in the 11.7–45.5 K range does not appreciably vary, within the experimental uncertainties (4%–7%). The largest difference in the individual $k(T)$ is around 20%, which is consistent with the difference between 10 and 50 K obtained in the computational study.

Figure 3 compares the experimentally measured and theoretically calculated rate coefficients for the $\text{CN} + \text{H}_2\text{S}$ reaction. The experimental data show a nearly temperature-independent behavior between 11.7 and 45.5 K, with rate coefficients around $4 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$. This behavior is fully consistent with the theoretical predictions, which indicate a weak negative temperature dependence over this range and rate coefficients of the same order of magnitude. The agreement within 20%–30%, well inside the combined experimental and theoretical uncertainties, supports a capture-controlled reaction mechanism governed by long-range interactions and submerged transition states. In particular, the absence of an activation barrier in the dominant pathway leading to HSCN, as predicted by the CCUS calculations, naturally explains the efficient reactivity observed at ultralow temperatures. Thus, the experimental results provide direct validation of the theoretical description of both the magnitude and temperature dependence of $k(T)$, justifying the use of the computed rate coefficients and branching assumptions in the astrochemical modeling discussed below.

4.2. Astrochemical Modeling

From the astrophysical point of view, the relative abundances of CN, H_2S , and HSCN were modeled using a pure gas-phase model. To evaluate the impact of the reported reaction rate coefficient on the chemical composition of cold interstellar clouds, we carried out chemical modeling

calculations adopting typical parameters of cold dark clouds, i.e., a kinetic temperature of 10 K, a H_2 volume density of 10^4 cm^{-3} , a cosmic-ray ionization rate of $1.3 \times 10^{-17} \text{ s}^{-1}$, a visual extinction of 30 mag, and the set of “low metal” elemental abundances (M. Agúndez & V. Wakelam 2013). We used the gas-phase chemical network from the latest release of the UMIST Database for Astrochemistry⁷ (T. J. Millar et al. 2024). The $\text{CN} + \text{H}_2\text{S}$ reaction is not included in astrochemical databases such as KIDA⁸ (V. Wakelam et al. 2024) or UMIST (T. J. Millar et al. 2024). As stated above, we included a rate coefficient of $4.0 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$ for the $\text{CN} + \text{H}_2\text{S}$ reaction, assuming that HSCN is the sole reaction product, as predicted by our calculations.

As shown in Figure 4, this reaction has no impact on the relative abundances of CN and H_2S , but it slightly impacts the modeled relative abundances of HSCN at evolving times between 10^5 and 10^6 yr. According to the model, the main reactions forming HSCN are the reactions of dissociative recombination with electrons of the cations HNCSH^+ , H_2NCS^+ , and H_2SCN^+ , where these species are formed upon protonation of the radical NCS followed by reaction with H_2 (J. Cernicharo et al. 2024), while the contribution of the $\text{CN} + \text{H}_2\text{S}$ reaction is 5–20 times less important than the ionic routes. We, however, note that the gas-phase chemical model severely underestimates the abundance of HSCN (observed in TCM-1; blue horizontal line in Figure 4) by about 3 orders of magnitude, which indicates that the model is missing efficient formation routes to this molecule. Moreover, the abundance of H_2S is also underestimated by the chemical model by a factor of ~ 400 with respect to the observations in TCM-1 (orange horizontal line in Figure 4), which means that the model severely underestimates the contribution of the $\text{CN} + \text{H}_2\text{S}$ reaction. This fact is of great interest because it means that the real contribution of the $\text{CN} + \text{H}_2\text{S}$ reaction to the formation of HSCN in cold dark clouds has to be boosted up with respect to the calculated one, which makes it likely that it becomes the main source of HSCN.

5. Conclusions

This work presents the first comprehensive study of the gas-phase reaction between CN and H_2S under conditions relevant to interstellar molecular clouds. Using the CRESU technique coupled with PLP-LIF, we experimentally determined the rate coefficient $k(T)$ between 11.7 and 45.5 K, finding values around $4 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$ with negligible temperature dependence. Complementary theoretical calculations employing AutoMeKin and CCUS models revealed that the dominant pathway leads to HSCN formation. Although the $\text{HCN} + \text{SH}$ channel is thermodynamically more exothermic, the HSCN pathway is kinetically favored, as it proceeds through a more deeply submerged transition state and exhibits a substantially lower Gibbs free energy of activation. These features, fully accounted for in the CVT/CCUS treatment, reflect both the lower effective barrier and the more favorable variational (transition-state tightness) effects of the HSCN channel. Astrochemical modeling indicates that this reaction, although not included in current astrochemical databases, can contribute to HSCN production in cold dark clouds, with an impact comparable to dissociative recombination routes. These

⁷ <https://umistdatabase.uk/>

⁸ <https://kida.astrochem-tools.org>

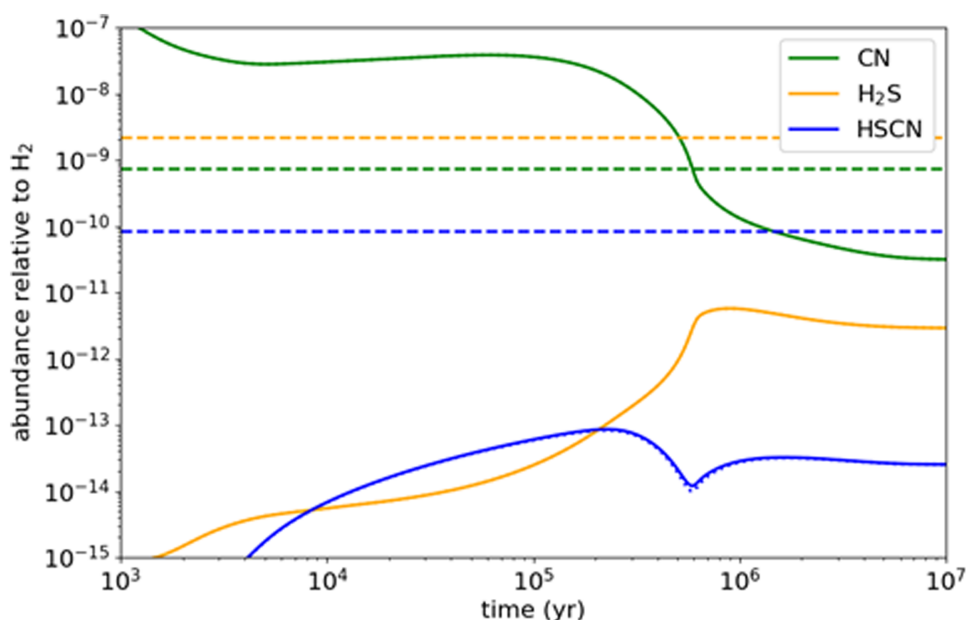


Figure 4. Abundances relative to H_2 calculated as a function of time for CN, H_2S , and HSCN. Dotted lines correspond to a model excluding the $\text{CN} + \text{H}_2\text{S}$ reaction while solid lines refer to the model in which the reaction is included with a rate coefficient of $4.0 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$. Horizontal dashed lines correspond to the abundances observed in the cold dark cloud TMC-1 (P. Pratap et al. 1997; D. Navarro-Almaida et al. 2020; J. Cernicharo et al. 2024).

findings highlight the importance of incorporating the $\text{CN} + \text{H}_2\text{S}$ reaction chemistry into astrochemical networks to better understand sulfur–nitrogen coupling and the origin of prebiotic molecules in the ISM.

Acknowledgments

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Author Contributions

M.A., F.M., and D.G. performed the experiments and data analysis; E.M.N., A.F.R., and A.L. carried out the theoretical

calculations; M.A. and J.C. ran the astrochemical model; E.J. designed and supervised the experiments, wrote the original draft of the paper, and got funds for carrying out the present experimental research. All authors contributed to the critical discussion and review of the draft.

Conflicts of interest

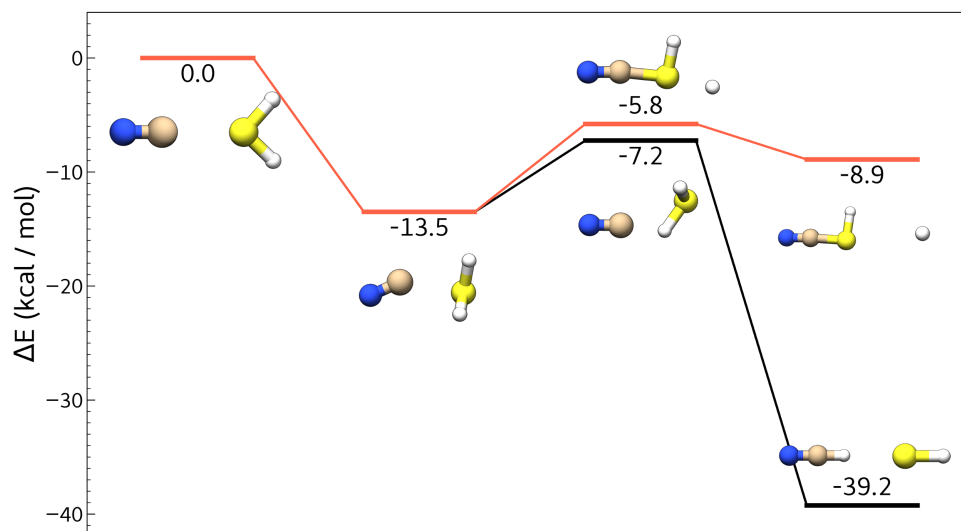
There are no conflicts to declare.

Appendix Supporting Information

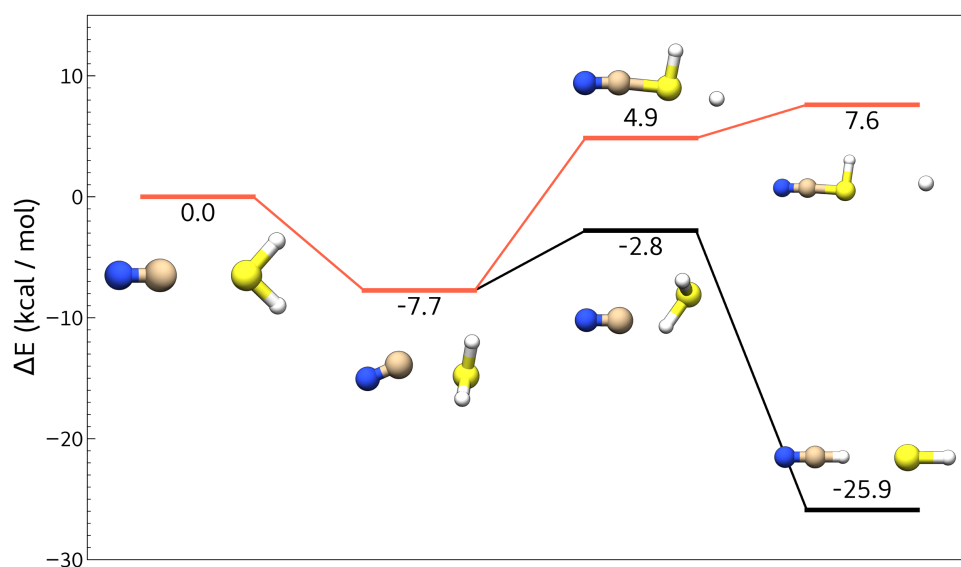
A.1. Benchmarking of DFT Methods Against Level3

To identify a computationally affordable method for evaluating variational effects, seven DFT methods were benchmarked against CCSD(T) reference calculations: B3LYP/6-31+G(d,p), mPWB1K/6-31+G(d,p), mPWB1K/MG3S, M08-HX/MG3S, M06-2X/6-31+G(d,p), M06-2X/cc-pVTZ, and M06-2X/MG3S. For each method, energy profiles including ZPVE corrections were constructed. Among these, M06-2X/MG3S was selected for the kinetic calculations, as it yielded barrier heights within $0.7 \text{ kcal mol}^{-1}$ (for the HCN channel) and $0.5 \text{ kcal mol}^{-1}$ (for the HSCN channel) of the CCSD(T) values. Figures A1–A7 display the corresponding energy profiles at the various DFT levels for reactions (R1a, black) and (R1e, red), including ZPVE corrections. The reactants ($\text{CN} + \text{H}_2\text{S}$), the van der Waals intermediate, transition states, and products are shown explicitly along each reaction pathway. The atomic color scheme used throughout is N = blue, C = beige, S = yellow, H = white.

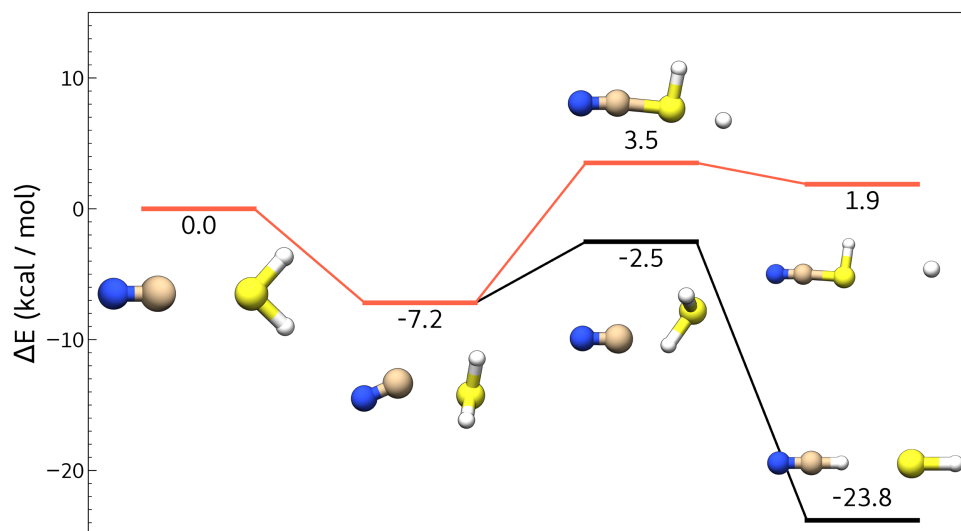
B3LYP / 6-31+G(d,p)

**Figure A1.** Energy profile including ZPVE correction for the B3LYP/6-31+G(d,p) method.

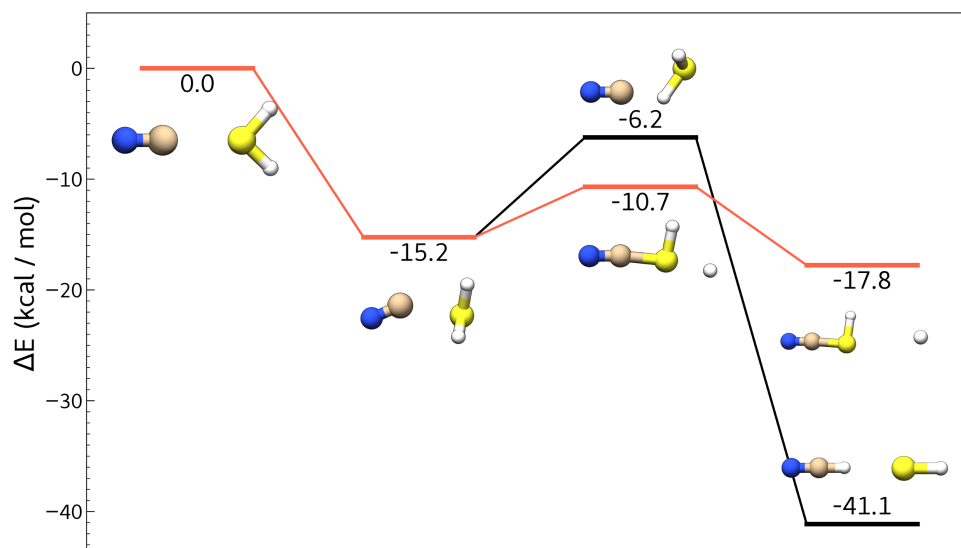
MPWB1K / 6-31+G(d,p)

**Figure A2.** Energy profile including ZPVE correction for the mPWB1K/6-31+G(d,p) method.

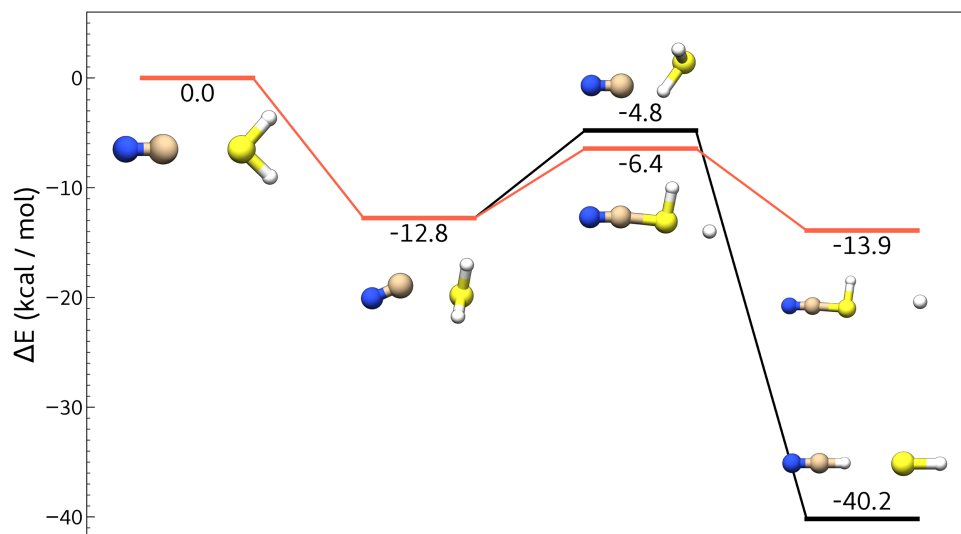
MPWB1K / MG3S

**Figure A3.** Energy profile including ZPVE correction for the mPWB1K/MG3S method.

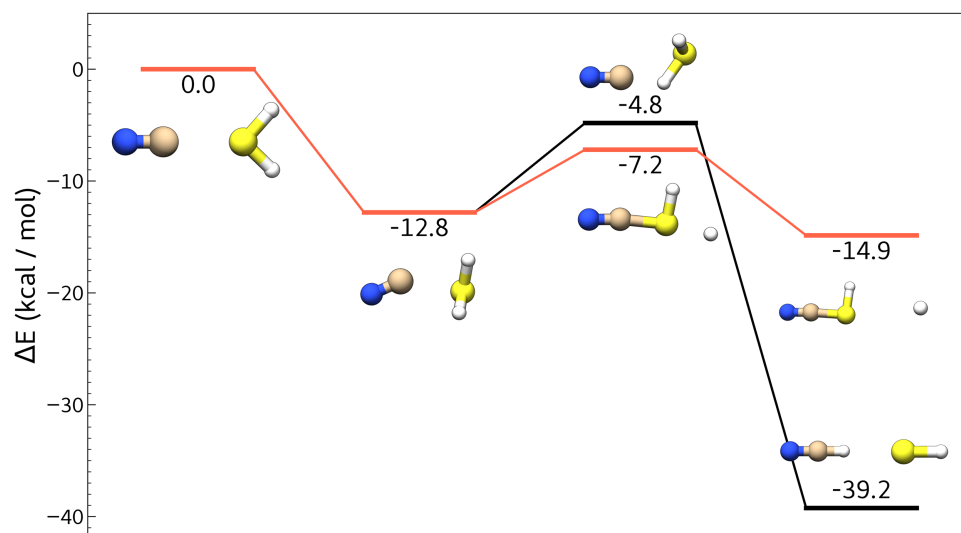
M08-HX / MG3S

**Figure A4.** Energy profile including ZPVE correction for the M08-HX/MG3S method.

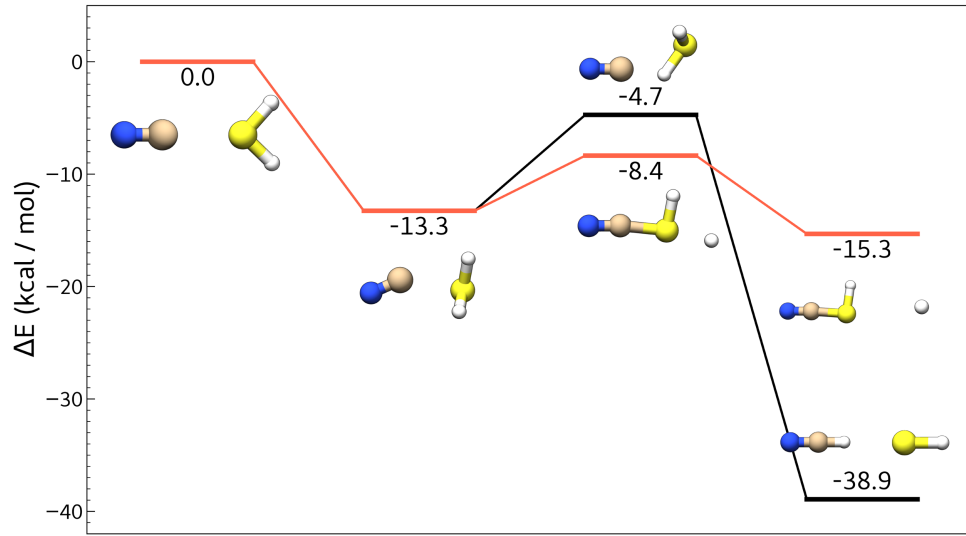
M06-2X / 6-31+G(d,p)

**Figure A5.** Energy profile including ZPVE correction for the M06-2X/6-31+G(d,p) method.

M06-2X / cc-pVTZ

**Figure A6.** Energy profile including ZPVE correction for the M06-2X/cc-pVTZ method.

M06-2X / MG3S

**Figure A7.** Energy profile including ZPVE correction for the M06-2X/MG3S method.*A.2. Computed Rate Coefficients*

The temperature-dependent rate coefficients obtained using TST and CVT are reported in Table A1, while the CCUS-derived rate coefficients for the individual channels and the overall reaction are summarized in Table A2.

Table A1
Rate Coefficients (in $\text{cm}^3 \text{s}^{-1}$) for Each Elementary Step, Computed Using TST and CVT

$T(\text{K})$	k^{assoc}	k_{1a}^{TST}	k_{1a}^{CVT}	k_{1e}^{TST}	k_{1e}^{CVT}
10	4.00×10^{-10}	1.52×10^{88}	2.63×10^{79}	2.35×10^{14}	1.32×10^{12}
20	3.56×10^{-10}	3.51×10^{38}	1.47×10^{34}	5.92×10^1	4.46×10^0
30	3.33×10^{-10}	7.70×10^{21}	9.35×10^{18}	2.93×10^{-3}	5.27×10^{-4}
40	3.17×10^{-10}	3.18×10^{13}	2.08×10^{11}	1.87×10^{-5}	5.23×10^{-6}
50	3.06×10^{-10}	2.76×10^8	4.95×10^6	8.65×10^{-7}	3.15×10^{-7}
60	2.96×10^{-10}	1.11×10^5	3.92×10^3	1.10×10^{-7}	4.77×10^{-8}
70	2.89×10^{-10}	4.07×10^2	2.32×10^1	2.49×10^{-8}	1.23×10^{-8}
80	2.83×10^{-10}	5.94×10^0	4.87×10^{-1}	8.20×10^{-9}	4.46×10^{-9}
90	2.77×10^{-10}	2.19×10^{-1}	2.38×10^{-2}	3.47×10^{-9}	2.03×10^{-9}
100	2.72×10^{-10}	1.55×10^{-2}	2.11×10^{-3}	1.75×10^{-9}	1.09×10^{-9}

Table A2
Rate Coefficients (in $\text{cm}^3 \text{s}^{-1}$) for Each Individual Reaction Channel and the Overall Process, Computed Using the CCUS Model

$T(\text{K})$	k^{CCUS}	k_{1a}^{CCUS}	k_{1e}^{CCUS}
10	4.00×10^{-10}	4.00×10^{-10}	2.00×10^{-77}
20	3.56×10^{-10}	3.56×10^{-10}	1.08×10^{-43}
30	3.33×10^{-10}	3.33×10^{-10}	1.88×10^{-32}
40	3.17×10^{-10}	3.17×10^{-10}	7.98×10^{-27}
50	3.06×10^{-10}	3.06×10^{-10}	1.95×10^{-23}
60	2.96×10^{-10}	2.96×10^{-10}	3.60×10^{-21}
70	2.89×10^{-10}	2.89×10^{-10}	1.53×10^{-19}
80	2.83×10^{-10}	2.83×10^{-10}	2.59×10^{-18}
90	2.77×10^{-10}	2.77×10^{-10}	2.36×10^{-17}
100	2.72×10^{-10}	2.72×10^{-10}	1.40×10^{-16}

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